

Chunlei Li (李春蕾)

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Summary

I'm a PhD candidate at the VR Lab of Beihang University from 2021, supervised by Prof. Qiping Zhao and Prof. Yang Gao. My research focuses on physical simulation in computer graphics. Currently I am a visiting student at University College London, supervised by Prof. He Wang. I am about to graduate in 2026/2027. Interest: Numerical Methods, Physical Simulation, Physical Based Animation, Reinforcement Learning, Computer Graphics, AI4S, PDE.

Education

University College London, Visiting PhD in Computer Science	Sept 2025 – Sept 2026
Beihang University (BUAA), PhD in Computer Science	Sept 2021 – Dec 2026
Beihang University (BUAA), MSc in Power Engineering and Engineering Thermophysics	Sept 2018 – June 2021
University of Michigan, Dearborn, Exchange Student in Energy and Power Engineering	Sept 2016 – May 2017
North China Electric Power University (Beijing), BSc in Energy and Power Engineering	Sept 2014 – July 2018

Publications

MGPBD: A Multigrid Accelerated Global XPBD Solver Chunlei Li, Peng Yu, Tiantian Liu, Siyuan Yu, Yuting Xiao, Shuai Li, Aimin Hao, Yang Gao, Qiping Zhao 10.1145/3721238.3730720 (SIGGRAPH)	Aug 2025
Parallel Constraint Graph Partitioning and Coloring for Realtime Soft-Body Cutting Peng Yu, Ruiqi Wang, Chunlei Li, Yuxuan Li, Xiao Zhai, Yuanbo He, Hongyu Wu, Aimin Hao 10.2312/pg.20251267 (Pacific Graphics)	Apr 2025
A Unified Particle-Based Solver for non-Newtonian Behaviors Simulation Chunlei Li, Yang Gao, Jiayi He, Tianwei Cheng, Shuai Li, Aimin Hao 10.1109/TVCG.2023.3341453 (IEEE Transactions on Visualization and Computer Graphics)	Dec 2023
Comparison between Two Eulerian-Lagrangian Methods: CFD-DEM & MPPIC on the biomass gasification in a fluidized bed Chunlei Li, Qitai Eri 10.1007/s13399-021-01384-2 (Biomass Conversion and Biorefinery)	Feb 2021
Comparative Study of Three Modified sCO₂ Brayton Recompression Cycles Based on Energy and Exergy Analysis with GA Optimization Chunlei Li, Qitai Eri 10.1504/IJEX.2021.115652 (International Journal of Exergy)	Jan 2021
Multi-objective Optimization of sCO₂, sCO₂/tCO₂ Cycles Based on Energy-Exergy-Economy Balanced Analysis Chunlei Li, Qitai Eri 10.1504/IJEX.2022.122308 (International Journal of Exergy)	Apr 2022

Experience

R&D, Zeno Tech – Online	June 2022 – Dec 2022
• Intern: R&D of the PBD method in the DCC software using C++.	

R&D, Taichi Graphics – Beijing

Feb 2023 – Sept 2023

- Intern: R&D of the PBD method. Supervised by Dr. Tiantian Liu, development of PBD solver.

R&D, Alibaba – Beijing

May 2025 – Nov 2025

- R&D of the multigrid accelerated GPU-based muscle simulation in Houdini.

Awards

Top 10 Outstanding Graduate Students Candidate of Beihang University: Beihang University

Outstanding Graduate of Beihang University: Beihang University

Outstanding Graduate of NCEPU (Beijing): NCEPU (Beijing)

Skills

Languages: CET Band 6: 578, CET Band 4: 560, TOEFL: 97, GRE: 323+3.5

DCC Software: Houdini(5 years user)

Physics Engines: Newton (PR contributor), MuJoCo, PhysX